

FENT2321

Wind Energy and Design of a Wind Turbine

Assignment 6: Reflecting on the generator design

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Problem 1: Operating point variables

Parameter	Value	Unit
Design RPM	1420	RPM
Induced voltage at design RPM	18.9	V
Terminal voltage at design OP	13.9	V
Output power at design OP	48.5	W
Efficiency at design OP	27.4	%

Problem 2: Changing the active length (Lactive, C26)

2 a) The induced voltage is doubled as the active length is a factor in the formula for induced voltage, hence doubling the active length equals to multiplying the induced voltage by 2.

2 b) The stator resistance is increased. This is because the total length of wire is increased as we double the active length of the coils which makes up for a lot of length of each coil.

2 c) Doubling the active coil length means making the part of the coil that is exposed to the magnetic field twice as long. In practice, the stator/rotor would typically be made longer in the axial direction.

2 d) If the "active length" of the magnets doesn't change the length of the coils that are added, will not experience the same B-field and thus not directly "double" the induced voltage. Leakage flux will also play a role near the end of the coils.

2 e) The formula for induced voltage becomes invalid now. We can't just double the active coil length because that implies that the magnetic field B is uniform and constant over the whole length. In the excel-sheet, the length of the magnets are stated to be 120mm, and will therefore not reach and supply the same field strength to the whole coil lengths.

Problem 3: Changing the number of turns (Nturn, C25)

3 a)

Nturn	Efficiency
20	12.8%
30	21.0%
40	25.1%
50	27.6%
60	29.3%
70	30.4%
80	31.3%
90	32.0%
100	32.6%

Figure 1: Efficiency for different numbers of turns.

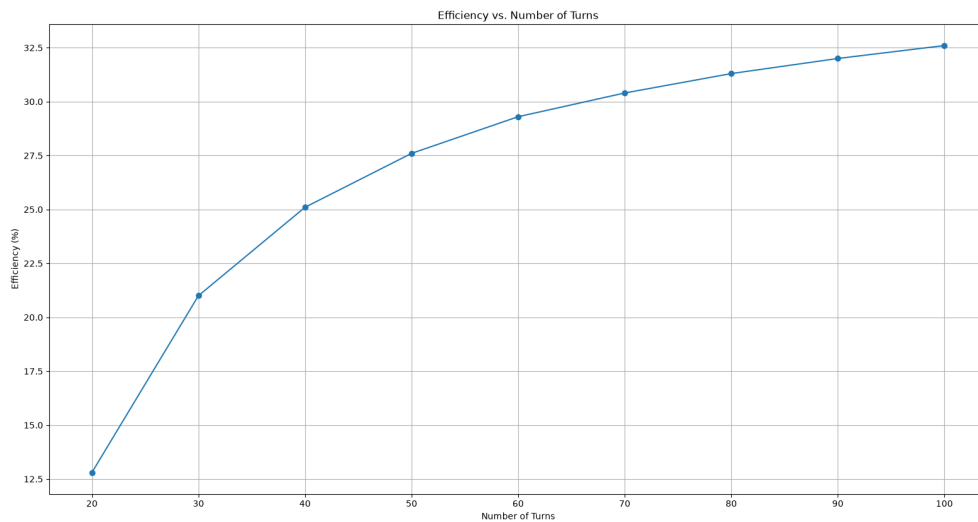


Figure 2: Your figure caption.

The efficiency is trending upwards for increased number of turns as is expected, but we can see the the drop off in efficiency increase which indicates a lower gain for increased number of turns as we go upwards.

3 b) With 100 turns, the coil would contain a large amount of copper wire compared with the original design. The coil would therefore take up more space and be more difficult to manufacture and wind. Keeping the same wire diameter also means that the coil becomes physically larger as more turns are added.

3 c) Increasing the number of turns affects several parameters in the Excel sheet. Two important examples are the stator resistance and the stator inductance. The stator resistance increases because the total length of copper wire increases with the number of turns. The stator inductance also increases strongly because the inductance in the Excel model is proportional to the square of the number of turns. Therefore, increasing the number of turns from 50 to 100 increases the inductance by a factor of four.

3 d) By increasing to 4 pole and 4 coils in series, in addition to choosing a copper diameter of 0.00118m, the efficiency initially increases to 62%, but this gives a T_{diff} of about 2Nm. By tuning R_{load} , I will not get to within the boundaries set without going over 50Ω which is above the upper boundary set in the excel. If this was possible however, our generator would be almost double the efficiency for $N_{turn} = 100$ in a).

Problem 4: Changing the diameter of the copper wire (Dcu, C24)

4 a) When the copper wire diameter is increased to 2.5mm, the cross-sectional area of the copper wire increases and a larger cross-sectional area gives a lower resistance. The lower resistance also reduces the copper losses and as a result, less power is lost as heat in the copper and the efficiency increases. In the excel model, the efficiency increases from the original value to approximately 41.5% when the wire diameter is increased to 2.5mm and the load resistance is adjusted to 17.25Ω so that the torque difference becomes < 0.01 .

4 b) Using thicker copper wire makes each turn physically larger because the wire occupies more space. This can increase the size of the coil and may require changes to the coil geometry. The thicker wire also reduces the resistance significantly. However, the effect on the overall generator geometry depends on whether there is enough available space for the larger wire. Therefore, using thicker wire can improve the electrical performance, but it can also create practical manufacturing and space constraints.

PART 2

My systems   

☐   PMSM 3 gear  Updated: 1.10.2026, 23:45:52	Price, EUR: 504863	Efficiency @100% load: 75.28	Efficiency @75% load: 75.1	Efficiency @50% load: 73.03	Efficiency @25% load: 71.5	Volume, m3: 53.0813	Footprint, m2: 32.7279	Weight, kg: 30727.62
THD(u), %: 94	THD(i), %: 4							

☐   DFIM 2 gear  Updated: 1.10.2026, 23:45:18	Price, EUR: 483782	Efficiency @100% load: 77.04	Efficiency @75% load: 77.24	Efficiency @50% load: 76.12	Efficiency @25% load: 72.22	Volume, m3: 35.1017	Footprint, m2: 20.4711	Weight, kg: 34769.03
THD(u), %: 27	THD(i), %: 4							

☐   PMSM 2 gear  Updated: 1.10.2026, 23:41:43	Price, EUR: 504863	Efficiency @100% load: 75.28	Efficiency @75% load: 75.1	Efficiency @50% load: 73.03	Efficiency @25% load: 71.5	Volume, m3: 53.0813	Footprint, m2: 32.7279	Weight, kg: 30727.62
THD(u), %: 94	THD(i), %: 4							

Figure 3: Comparison of different drive trains.